

Scheduling Algorithms

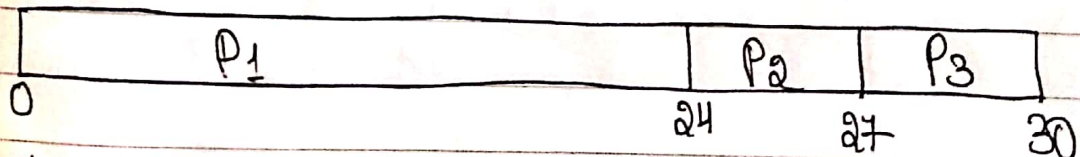
1. First-Come, First-Served Scheduling

Problems

1. Consider the following set of processes that arrive at time 0, with the length of the CPU burst given in milliseconds

<u>Process</u>	<u>Burst Time</u>
P ₁	24
P ₂	3
P ₃	3

If the processes arrive in the order P₁, P₂, P₃ and are served in FCFS order, we get the result shown in the following Gantt chart:

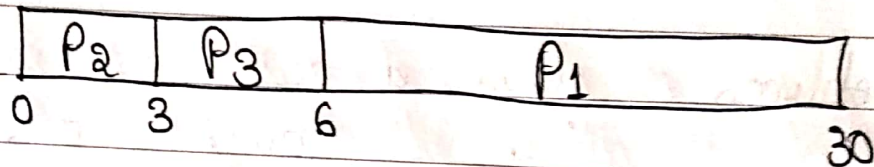


The Average waiting time is (AWT)

$$P_1 + P_2 + P_3$$

$$AWT = (0 + 24 + 27) / 3 = 17 \text{ milliseconds}$$

If the process arrive in the order P_2, P_3, P_1 , the result will be shown in the following Gantt chart



Average waiting Time

$$\begin{array}{ccc} P_1 & P_2 & P_3 \\ (6 + 0 + 3) / 3 = 3 \text{ milliseconds} \end{array}$$

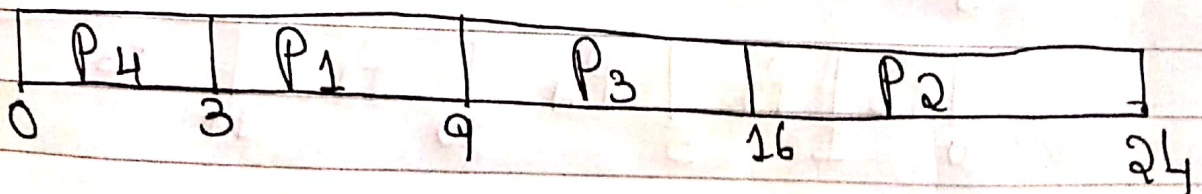
2. Shortest - Job - First Scheduling

Problems - Non - Preemptive

1. Consider the following set of processes, with the length of the CPU burst given in milliseconds

<u>Process</u>	<u>Burst Time</u>
P ₁	6
P ₂	8
P ₃	7
P ₄	3

Using SJF scheduling, we would schedule process according to the following Gantt chart



Average Waiting Time

$$\frac{P_1 \quad P_2 \quad P_3 \quad P_4}{(3 + 16 + 9 + 0) / 4} = 7 \text{ millisecond}$$

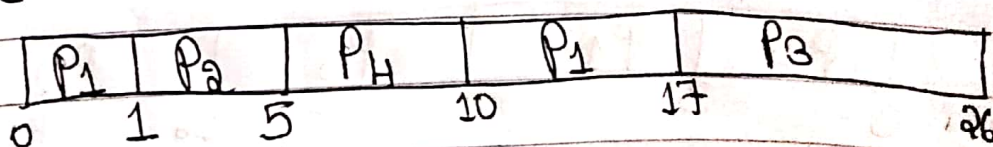
Shortest Remaining time first Schedu

Problems - Preemptive

1. As an example, consider the following processes, with the length of the CPU burst given in milliseconds.

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P ₁	0	8 - 1 = 7
P ₂	1	4 - 1 = 3
P ₃	2	9
P ₄	3	5

The resulting preemptive STF schedule is as depicted in the following Gantt chart:



Average waiting time is

$$\begin{aligned}
 &= \frac{(P_1 + P_2 + P_3 + P_4)}{4} \\
 &= \frac{(10 - 1) + (1 - 1) + (17 - 2) + (5 - 3)}{4} \\
 &= \frac{26}{4} = 6.5 \text{ milliseconds}
 \end{aligned}$$

Process	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
(P)	(AT)	(BT)	(CT)	(TAT)	(WT)
P ₁	0	8	17	17	9
P ₂	1	4	5	4	0
P ₃	2	9	26	24	15
P ₄	3	5	10	7	2

Calculate Turn Around Time (TAT)

$$TAT = \text{Completion Time} - \text{Arrival Time}$$

$$TAT = CT - AT$$

$$(TAT)$$

P ₁	17	- 0	= 17
P ₂	5	- 1	= 4
P ₃	26	- 2	= 24
P ₄	10	- 3	= 7

Average Turn around Time (A-TAT)

$$= \frac{P_1 + P_2 + P_3 + P_4}{4}$$

$$= \frac{17 + 4 + 24 + 7}{4}$$

$$= 52 / 4$$

$$= 13 \text{ millisecond}$$

Calculate Waiting Time (WT)

$$WT = \text{Turn around Time} - \text{Burst Time}$$
$$TAT - BT$$

$$\begin{array}{l} P_1 = 17 - 8 = 9 \\ P_2 = 5 - 4 = 1 \\ P_3 = 24 - 9 = 15 \\ P_4 = 7 - 5 = 2 \end{array}$$

Average Waiting Time (AWT)

$P_1 \quad P_2 \quad P_3 \quad P_4$

$$\begin{aligned} AWT &= (9 + 1 + 15 + 2) / 4 \\ &= 26 / 4 \\ &= 6.5 \text{ milliseconds} \end{aligned}$$

3. Priority Scheduling

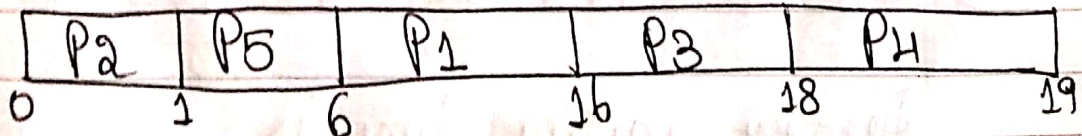
Problems

1. ~~Discuss~~ Consider the following set of processes, assumed to have arrived at time 0, with the length of the CPU burst given in milliseconds

Process Burst time Priority

P ₁	10	3
P ₂	1	1
P ₃	2	4
P ₄	1	5
P ₅	5	2

Using priority scheduling, we would schedule these processes according to the following Gantt chart:-



Average Waiting Time is

$$= \frac{P_1 + P_2 + P_3 + P_4 + P_5}{5}$$

$$= \frac{(6 + 0 + 16 + 18 + 1)}{5}$$

$$= 41 / 5 = 8.2 \text{ milliseconds}$$

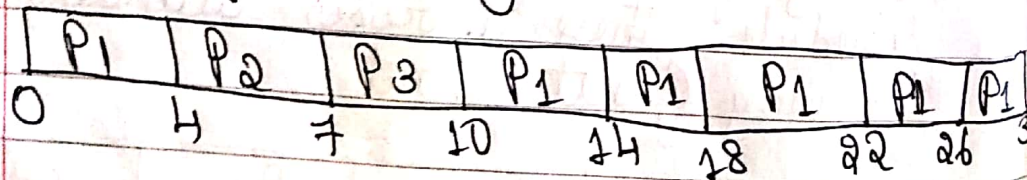
4. Round - Robin Scheduling

Problems -

1. Consider the following set of processes that arrives at time 0, with the length of the CPU burst given in milliseconds. Time Quantum = 4

<u>Process</u>	<u>Burst Time</u>
P ₁	24
P ₂	3
P ₃	3

Using Round Robin scheduling, we would schedule these processes according to the following Gantt chart.



Average waiting time is

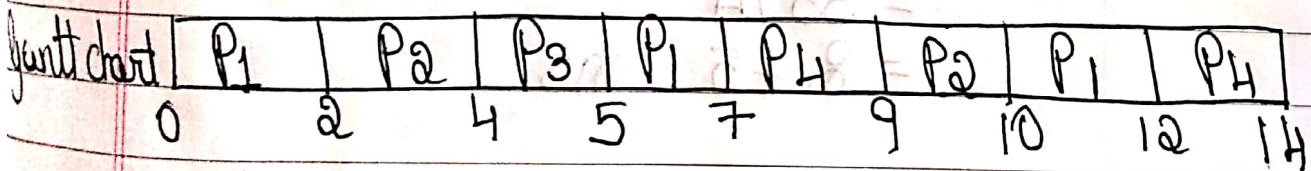
$$\begin{aligned}
 &= \frac{P_1 + P_2 + P_3}{3} \\
 &= \frac{(7 + 4 + 6)}{3} \\
 &= 5.66 \text{ milliseconds}
 \end{aligned}$$

2. Consider the following set of processes with CPU burst time (in ms)

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P ₁	0	$6 - 2 = 4 - 2 = 2$
P ₂	1	$3 - 2 = 1$
P ₃	2	1
P ₄	3	$4 - 2 = 2$

Compute the waiting time and average turnaround time for the above process using RR (time quantum = 2 ms) scheduling algorithm.

Ready Q ~~P₁~~ ~~P₂~~ ~~P₃~~ ~~P₁~~ ~~P₄~~ ~~P₂~~ ~~P₁~~ ~~P₄~~



Process	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
P ₁	0	6	12	12	6
P ₂	1	3	10	9	6
P ₃	2	1	5	3	2
P ₄	3	4	14	11	7

Average Turn Around Time

$$TAT = CT - AT$$

$$P_1 = 12 - 0 = 12$$

$$P_2 = 10 - 1 = 9$$

$$P_3 = 5 - 2 = 3$$

$$P_4 = 14 - 3 = 11$$

$$\begin{aligned} \text{A-TAT}_{\text{avg}} &= (12 + 9 + 3 + 11) / 4 \\ &= 35 / 4 \\ &= 8.75 \text{ ms} \end{aligned}$$

Waiting Time = Turn around - Burst time

$$P_1 = 12 - 6 = 6$$

$$P_2 = 9 - 3 = 6$$

$$P_3 = 3 - 1 = 2$$

$$P_4 = 11 - 4 = 7$$

$$\begin{aligned} \text{AWT} &= (6 + 6 + 2 + 7) / 4 \\ &= 21 / 4 \\ &= 5.25 \text{ ms} \end{aligned}$$

Examples - Need to Solve by Students

1. Calculate the average waiting time by drawing Gantt chart using FCFS (First Come First Serve), SRTF (Shortest Remaining Time First), RR (Round Robin) [$q = 2$ ms] algorithms

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P ₁	0	9
P ₂	1	4
P ₃	2	9
P ₄	3	5
P₅		